Relate Fractions, Decimals, and Percents

Lesson 4-2

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

25% means 25 out of 100

Percent

A way to compare a number to 100, shown with the % sign.

0.75 means seventy-five hundredths

Decimal

A number with a dot, like 0.5, that shows a part.

1/2 = 0.5 = 50%

Equivalent

Having the same value, just written a different way.

Common benchmarks: 25%, 50%, 75%

Benchmark

A familiar number you use to guess, like 50% or 1/2.

$40 shirt at 25% off: discount = $10

Discount

Money taken off the first price to make it cheaper.

Key Ideas & Notes

- You're designing the scoring system for a new arcade game.
- Players earn points as fractions of the total possible score.
- The leaderboard shows scores as percentages, but the game engine uses decimals.
- You need to convert between all three forms to build the display!
- Complete the conversion table. Each row shows the same value in three forms.

Think About It

- What are three different ways to write the same number?
- If a player scored 3/4 of the points, what percent is that?
- Are some fractions easier to convert than others?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A player scored 3/4 of the points. The leaderboard shows percents and the game engine uses decimals. What are three different ways to write the same score?

Level 1 support

Start here: 3/4, 0.75, and 75% all name the same score.

The fraction ____ is the same as the decimal ____ and the percent ____.

La fracción ____ es igual al decimal ____ y al porcentaje ____.

I can write the same value as a fraction, a ____, or a ____.

Puedo escribir el mismo valor como fracción, ____ o ____.

Word bank: percent | decimal | equivalent | benchmark

Listen for: *Students give 3/4 = 0.75 = 75% and recognize these are three forms of one value, not three different scores.*

Level 2

Which form would be easiest to read on a leaderboard, and which is easiest for the game engine to calculate with? Justify your choices.

- The ____ form is easiest to read because ____.
- The ____ form is easiest to calculate with because ____.

EXPLORE

How did you convert a fraction like 3/5 into a decimal and a percent in the table?

Level 1 support

Start here: I divided the numerator by the denominator, then multiplied by 100 to get the percent.

To change ___ to a decimal I divided ___ by ___.

Para cambiar ___ a decimal dividí ___ entre ___.

To get the percent I multiplied the decimal by ___.

Para obtener el porcentaje multipliqué el decimal por ___.

Word bank: **percent** **decimal** **equivalent** **benchmark**

Listen for: *Students describe dividing numerator by denominator (3 div 5 = 0.6) and multiplying by 100 to get 60%, and call the three forms equivalent.*

Level 2

Some fractions like 1/4 give a clean decimal, but others do not. How can a benchmark like 1/2 or 1/4 help you estimate before you divide?

- A benchmark helps because ___.
- I know ___ is close to ___ because ___.

CONNECT

A sale says 'Everything 3/5 off!' One friend says that is 35% off and another says 60% off. Who is right, and how do you know?

Level 1 support

Start here: 3/5 equals 60%, so the friend who said 60% off is correct.

The friend who said ___% is correct because 3/5 = ___ = ___%.

El amigo que dijo ___% tiene razón porque 3/5 = ___ = ___%.

The other friend is wrong because ___.

El otro amigo está equivocado porque ___.

Word bank: **percent** **decimal** **equivalent** **discount**

Listen for: *Students convert 3/5 to 0.6 and 60%, identify 60% off as correct, and can find the $40 game becomes $16.*

Level 2

The other friend got 35% by reading the digits 3 and 5. Explain why that shortcut fails, and describe a discount where reading the digits would accidentally give the right percent.

- Reading the digits fails because ___.
- It would only work by accident when ___.

Guided Examples

Example 1

Convert 3/5 to a percent.

Solution: 3 ÷ 5 = 0.6 = 60%.

Answer: A. 60%

Example 2

Convert 7/20 to a percent.

Solution: 7 ÷ 20 = 0.35 = 35%.

Answer: A. 35%

Example 3

Which decimal is equivalent to 45%?

Solution: To convert a percent to a decimal, divide by 100: 45 ÷ 100 = 0.45.

Answer: A. 0.45

Write About the Math

The Writing Revolution

I can explain my conversions using the words percent, decimal, equivalent, and benchmark.

1. Kernel Sentence

subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.
Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.
Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Percent matters in math
Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Percent matters in math because ____.** *Porcentaje importa en matemáticas porque ____.*
but *pero*	**Percent matters in math, but ____.** *Porcentaje importa en matemáticas, pero ____.*
so *entonces*	**Percent matters in math, so ____.** *Porcentaje importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question
Pregunta

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The fraction ____ is the same as the decimal ____ and the percent ____.
La fracción ____ es igual al decimal ____ y al porcentaje ____.

I can write the same value as a fraction, a ____, or a ____.
Puedo escribir el mismo valor como fracción, ____ o ____.

To change ____ to a decimal I divided ____ by ____.
Para cambiar ____ a decimal dividí ____ entre ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What fraction is equal to 0.75?

A. 3/4
B. 7/5
C. 3/5
D. 7/10

Show your work:

2. A game shows your score as 7/8 complete. What percent is that?

A. 87.5%
B. 78%
C. 75%
D. 88%

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says 3/8 is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

Sentence starter: The student is ____ because 3/8 = ____ = ____%, which is ____ than 40%.

Show your work:

Reflect — Exit Ticket

Which shows 0.45 as a fraction and a percent?

A. 9/20 and 45%

B. 45/10 and 4.5%

C. 4/5 and 45%

D. 9/20 and 4.5%

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 3/4 *— 0.75 = 75/100. Simplify by dividing both by 25: 75÷25 = 3 and 100÷25 = 4. So 0.75 = 3/4.*
2. **Try It 2:** A. 87.5% *— 7 ÷ 8 = 0.875, and 0.875 × 100 = 87.5%.*
3. **Exit Ticket:** A. 9/20 and 45% *— 0.45 = 45/100 = 9/20 (simplified). 0.45 × 100 = 45%.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).